

# Adaptation Experiment 3: Learned Context Gates

Thesis experiment note

## 1 Theoretical overview and goal

Retrieval can help some queries and harm others. A gate chooses between the vanilla forecast  $\hat{y}_0$  and the context forecast  $\hat{y}_c$ . On  $T_2$  the supervision signal is the loss advantage

$$\Delta_h = (y_h - \hat{y}_{0,h})^2 - (y_h - \hat{y}_{c,h})^2,$$

so  $\Delta_h > 0$  means context is preferable. Scalar gates make one decision per window; horizon gates make one decision per forecast step. Regressors estimate  $\Delta$ , while classifiers estimate the event  $\Delta > 0$ .

**Experimental goal.** Learn when retrieval should be trusted, quantify the value of query-level features, and compare attainable gates with target-aware oracle upper bounds.

## 2 Implementation details

The gate features summarize the context–vanilla difference, query mean and standard deviation, neighbor look-back statistics, same-user ratio, neighbor age, distance-weight concentration, and average distance. Horizon gates also receive the full horizon-wise context difference.

| Gate                | Training target          | Decision granularity |
|---------------------|--------------------------|----------------------|
| Bayes no-feature    | Mean $T_2$ advantage     | Scalar / horizon     |
| CatBoost classifier | $\mathbb{I}[\Delta > 0]$ | Scalar / horizon     |
| CatBoost regressor  | $\Delta$                 | Scalar / horizon     |
| Oracle              | True target loss         | Scalar / horizon     |

CatBoost uses 300 iterations, learning rate 0.03, depth 4, balanced classes, and seed 1. Models are trained only on  $T_2$  and applied without refitting to  $T_3$ . The retrieval grid is raw/instance Euclidean distance with  $k \in \{1, 3, 10\}$  for Electricity and Solar at 168–24 and 672–168.

**Launcher:** `src/adaptors/gates/slurm/run.slurm`

**Active module:** `src.adaptors.baselines.evaluate -family gates`

**Standalone module:** `src/adaptors/gates/evaluate.py`

Outputs include per-method metrics, prediction payloads, gate diagnostics, and feature importance plots.

## 3 Interpretation

The oracle gap measures headroom, not deployable performance. The comparison between the no-feature Bayes gate and feature-based gates isolates the benefit of conditional decisions. Final claims must use  $T_3$  and should report both mean error and user-tail error because a gate can improve the mean while systematically rejecting useful context for a subgroup.